Engineering Mathematics - I · Unit_05 · MATLAB

# DRAFT — UE26MA141A · Unit_05 · MATLAB · Video

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

# · How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

| | |
|---|---|
| Source text | Verbatim or OCR-derived text, kept separate from editorial notes. |
| Candidate | Automated topic/source association requiring manual confirmation. |
| Low ASR confidence | Transcript segment requires audio verification. |
| DRAFT | Human review remains incomplete. |

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M1

# Video

## 1 Video

Study focus: Video. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000082 · EV0000083

SHORT-NOTE RECAP

**Core explanation · 00:00–04:59**

In this MATLAB session, we explore partial derivatives using MATLAB.

**Key details**

- And then after that we need to define the function.
- And then the partial derivative of the function f with respect to x is written as dif.
- of f comma x and with respect to y we write it as dif f of f comma y.
- And then the function f is equal to sine of x.
- plus y cube, we're write it as y cap three plus x power ten that is x cap ten minus y square plus log of x.
- If f is equal to x square plus 2y square minus 22, we need to find fxx and f y.
- The symbolic variables x and y are defined and f is equal to the function x square plus 2y square minus 22 is defined and then dif f of f x2 is nothing but partial derivative of f with respect to x and again also with respect to x.
- If f is equal to x y q plus tan x plus cos of root log x then find dfx here the.
- If f is equal to xy cube divided by x plus y, then find fx f y fx f x y and f yx.
- We know that the code to find the partial derivative of.
- And dif of f comma x comma y is nothing but partial derivative of f with respect to x first and then with respect to y.
- And at last dif of f x is nothing but partial derivative of f with respect to y first and then with respect to x.

**Example or application from the lecture**

- If f is equal to sine of x plus y cube plus x to the power 10 minus y square plus log of x as a given function, then find fx and f y.

- Here we need to find the partial derivative of the given function f with respect to x and with respect to y also.
- We'll move to the next example.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

**Core explanation · 05:00–09:59**

If u is equal to x square y plus y square z plus z square x, we need to find the first order partial derivative of u.

**Key details**

- Dif of u x.
- If u is equal to root of x square plus y square plus z square plus tan inverse of x y, find the first order partial derivative of u.
- We are defining.
- It is nothing but tan inverse is written as a tan of x into y.
- A partial derivative first order partial derivative of u with respect to x y z r also written as d ff of u x1 d ff of u y one df of u z1 respectively.
- And this is the output we are obtaining for this problem.
- When all the second order second partial derivatives of f of x y is equal to log of one by x minus one by y at one comma two.
- Partial second order second partial derivatives.
- And here we are initially defining.
- A of x y as dif of f of x2, partial derivative of f with respect to x twice, and partial derivative of f with respect to y twice as b of x comma y.
- And C of x comma y is nothing but partial derivative of f with respect to x first and then with respect to y.
- D of x y is nothing but partial derivative of uf with respect to y, comma.
- X is nothing but derivative of f with respect to y first and then with respect to x.
- And also we are continuing the same code with a of 1 comma 2, b of 1 comma 2, and c of 1 comma 2 d of 1 comma 2 because at upper point 1 comma 2 we need to find the second partial derivatives of the function.
- And this is the output we obtained.
- And at 1 comma 2 we are obtaining 0 minus 3 by 4, 1 and 1.
- If u is equal to e power x y plus y z plus zx, then find u x y uyz and uz.
- Dif of u y z is nothing but partial derivative of u with respect to y first and then with respect to z.
- DIFF of U Zx is nothing but partial derivative of U with respect to Z first and then with respect to X.
- If f is equal to e power x y z then find f x z f y z so symbolic variables x y z are defined and also the even function is defined in the second step.
- So d f of x f x y z is nothing but partial derivative of f with respect to x first and then with respect to y and at the last with respect to z.

- In the similar way, f xs z is nothing but partial derivative of f with respect to x first and again also with respect to x and at last with respect to z.
- The f of f y z is nothing but partial derivative of uf with respect to.
- Y first and again with respect to y and at last with respect to z.
- Next, we'll move on to the Telia series expansion of the function of single variable.

**Example or application from the lecture**

- So we need to define all the three symbolic variables s y s xy z and then the given function is defined.
- one is nothing but partial derivative of the given function u with respect to x and u y and u z is also done in a similar way and this is the output we are obtaining.
- If z is equal to sin two x plus three y as a given function, we need to find the first order partial derivative of z.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

**Core explanation · 10:00–14:59**

We need to define the symbolic variables in the initial stage and also the function has to be defined.

**Key details**

- T is equal to tailor of f of f, comma.
- Expansion point to order 3.
- Expand the function about the point x is equal to 2.
- comma with above the point X is equal to 2, so we are defining it as expansion point.
- Two expansion point two.
- And with order.
- [ASR low confidence; check audio].
- Next function f of x is equal to log of cos x about the point x is equal to pi by 3 up to 5th degree terms.
- It has also worked in a similar way.
- Here the expansion point is.
- Pi by 3 and order it is up to 5th degree terms, so we are taking order 6.
- Expand f of x is equal to log of secant x about the origin up to six degree terms.
- S y must x and also after that the function log of secant x is defined.
- T is equal to tailor of f with the order seven.
- Here it is about the origin, so no need to mention the expansion point.
- It is up to six degree terms, so we are writing the order seven.
- f of x is equal to sine of log of x square plus 2x plus 1 about the origin up to 6 degree terms.
- So same way x is defined and the function f is defined and also it is also about the origin.
- Is the code we are writing.
- That is about the origin.

- expand f of x comma y is equal to e power x into cos y above the point x is equal to 1 and y is equal to pi by 4 up to three degree terms.
- And the function is f is equal to exponent of x.
- into cos y.
- Taylor of the function yf.
- One comma [ASR low confidence; check audio].
- I buy sport.

**Example or application from the lecture**

- So the symbolic variable x is defined and the given function log of cos x is defined.
- The same way t is equal to tailor of the given function f.
- It is about the point XY that given it as.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

**Core explanation · 15:00–19:59**

I have given up to three degrees terms.

**Key details**

- So next problem is f of x y is equal to x cube plus y cube plus xy square about x is equal to 1 and y is equal to 2 up to 4th degree terms.
- It is also the similar problem to which we just now.
- Next function e power y into log of 1 plus x above the original to fourth degree terms.
- comma x y so it is about the origin so no need to mention the points over there and it is up to four degree terms so the order five and this is the result we are obtaining f of x y is equal to d power x into tan y above the origin up to fifth degree terms.
- That is about the origin no need to mention the points over there in the line in the statement and it is up to fifth degree terms so the order six and this is the output which we are obtaining.

**Example or application from the lecture**

- For this given problem.
- The function is also defined and here the points that are given at as 1 comma 2 and it has up to 4 degree terms.
- So symbolic variables are defined and also the given function is defined.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

# A Symbol glossary

| Symbol | Name | Meaning | Appears in |
|---|---|---|---|

# B Sources, coverage and gaps

| Source | Path | Use |
|---|---|---|
| Video transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/006_Video__bb969ddbec162496.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| Video transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/006_Video__bb969ddbec162496.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| Video transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/006_Video__bb969ddbec162496.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9998; only the JSON segmentation supplies the displayed timestamps. |
| Video transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/006_Video__bb969ddbec162496.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

### Known gaps, conflicts and corrections

1. 4 rendered pages in 001_Slides__7ac00f797de85ee4.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 002_Slides__ef919841425356a4.slides.md have no readable `.slides.md` text; their original page images are included.
3. 1 rendered pages in 003_Slides__3d7af4a88d5b4490.slides.md have no readable `.slides.md` text; their original page images are included.

4. 1 rendered pages in 004_Slides__4150cd4c33be9a2c.slides.md have no readable `.slides.md` text; their original page images are included.
5. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
6. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
7. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
8. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
9. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
10. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
11. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

## Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

| Source | Unit counts | Example decisions |
|---|---|---|
| SRC00018 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0063. Sidecar variants compare... |

| Source | Unit counts | Example decisions |
| --- | --- | --- |
| SRC00019 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0063. Sidecar variants compare... |
| SRC00020 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0063. Sidecar variants compare... |

| Source | Unit counts | Example decisions |
| --- | --- | --- |
| SRC00021 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0063. Sidecar variants compare... |

# C Revision and retrieval practice